

- Q.21 Write about total cycle of casuticiging process.  
 Q.22 Write any four physical properties of green liquor?

### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 With neat & clean diagram. Explain the working of cyclone evaporator.  
 Q.24 Write short note on any two:-  
 a) Multiple effect evaporator  
 b) Green liquor & white liquor  
 c) Oxidizers  
 Q.25 Make a flow sheet to explain black liquor recovery with its advantage and disadvantages.

No. of Printed Pages : 4  
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220663

### 6th Sem/ Chemical (Pulp & Paper) Subject : Chemical Recovery

Time : 3 Hrs.

M.M. : 60

### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which chemical is typically used to precipitate lignin from black liquor in a soda pulping process?  
 a) Sodium hydroxide    b) Sodium carbonate  
 c) Sulfuric acid        d) Calcium chloride  
 Q.2 In a Kraft pulping process, the main purpose of chemical recovery is to :  
 a) Recover heat from the black liquor  
 b) Recover chemicals from the black liquor  
 c) Reduce wastewater discharge  
 d) All of the above  
 Q.3 What is the main purpose of a multiple effect evaporate in chemical recovery?  
 a) To concentrate the black liquor  
 b) To cool the black liquor  
 c) To separate lignin from the black liquor  
 d) To recover heat from the black liquor

Q.4 Which of the following is a major environmental concern associated with chemical recovery processes?

- a) Air pollution from combustion of black liquor
- b) Water pollution from wastewater discharge
- c) Odor emissions from chemical processing
- d) All of the above

Q.5 What is the main purpose of the causticizing process in a Kraft pulping process?

- a) To convert sodium carbonate to sodium sulfide.
- b) To convert sodium sulfide to sodium hydroxide
- c) To convert sodium carbonate to sodium hydroxide
- d) To convert sodium hydroxide to sodium carbonate

Q.6 What is the full form of M.E.E.?

- a) Multiple Energy Evaporator
- b) Multiple Effect Evaporator
- c) Multiple Economy Evaporator
- d) Multiple Environment Evaporator

(2)

220663

## SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write formula of slaked lime?

Q.8 Define Sulfidity.

Q.9 Smelt is \_\_\_\_\_.

Q.10 Write one application of chemical recovery?

Q.11 Define Cauticizing efficiency.

Q.12 Dreg washers are used for \_\_\_\_\_.

## SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the Lime Kiln with diagram?

Q.14 Compare causticity and black liquid oxidizers.

Q.15 What is the effect of concentration of black liquor on evaporator?

Q.16 Discuss start up procedure of a multiple effect evaporator.

Q.17 Explain the Slacking process of green liquor.

Q.18 Write the physical characteristics of soda and sulfate black liquor in details.

Q.19 Describe about lime mud handling?

Q.20 Discuss the different challenges in the Kraft chemical recovery.

(3)

220663